Wavefunction Collapse as Retrodiction Failure: From Gravitational Decoherence to the AND–OR Transition

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Wavefunction collapse—the transition from quantum superposition to classical definiteness—is given a precise information-theoretic characterization: *collapse is Petz recovery failure* ($\tau \to 1$). Applying the temporal asymmetry parameter $\tau = 1 - F$ from the Petz recovery framework [S.-K. Huang, companion paper] to the gravitational dephasing channel of Pikovski et al., the full time-dependent collapse curve $\tau(t) = 1 - \exp(-N\Gamma t^2)$ is derived, with collapse and objectivity (quantum Darwinism redundancy $R \to N$) emerging at the same timescale. The Penrose–Diósi timescale $t_c = \hbar/E_G$ is recovered as a special case, with $\tau(t) \leq 1 - \exp(-E_G t/2\hbar)$ providing an upper bound on the full dynamics where Penrose gives only one time point. Three recent results sharpen the physical content: Liu, Bai, and Scarani show that the improper-to-proper mixture transition (AND→OR) changes the retrodictive structure itself—this transition *is* collapse, operationally defined; Jean, Silva, and Vilasini prove that $\tau = 0$ is operationally equivalent to post-selected closed timelike curves, giving τ the role of continuous interpolation between time-symmetric and time-directed regimes; Kuang, Torii, and Buscemi show that for measurement channels, all f-divergences yield the same retrodiction, making τ the unique irreversibility measure. The causal chain from non-uniform gravitational redshift to collapse is traced explicitly: c non-uniform $\to$ differential phase accumulation $\to \Sigma > 0 \to \tau > 0 \to$ AND→OR. Predictions for molecule interferometry, optomechanical, and gravitational entanglement experiments are tabulated. The universal bound $F_{\text{spatial}}(t) \geq \exp(-\Sigma_{\text{grav}}(t)/2)$ constrains any recovery protocol; deviations from the standard-QM baseline $\tau(t)$ would signal new physics.

INTRODUCTION

In a companion paper [1], we established that the Petz recovery map [2, 3] connects quantum erasure, Bayesian retrodiction, quantum error correction, and thermodynamic irreversibility. The temporal asymmetry parameter

$$\tau = 1 - F\big(\rho,\, \tilde{\mathcal{R}}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho))\big) \tag{1}$$

quantifies the degree to which a quantum channel $\mathcal{N}$ distinguishes past from future: $\tau = 0$ for closed (unitary) evolution, $\tau > 0$ for open systems, with the bound $\tau \leq 1 - \exp(-\Sigma/2)$ linking recovery failure to entropy production Σ [4, 5].

A central question in quantum foundations remains: when does a quantum superposition become a classical mixture? The decoherence program [6–8] identifies environment-induced decoherence as the mechanism, but leaves open what distinguishes "for all practical purposes" (FAPP) collapse from genuine irreversibility. Meanwhile, Penrose [9] and Diósi [10] have argued that gravitational self-energy sets a fundamental collapse timescale $t_c = \hbar/E_G$.

Three recent results transform the landscape. Liu, Bai, and Scarani [12] proved that proper and improper mixtures—though sharing the same density matrix—give different retrodictions via the Petz map. The decoherence transition from improper to proper mixture (AND→OR) is not merely a practical change but a qualitative shift in the retrodictive structure. Jean, Silva, and Vilasini [13] established that every time-symmetric quantum process is operationally equivalent to a post-selected closed timelike curve (P-CTC) circuit, and vice versa. Kuang, Torii, and Buscemi [14] showed that for measurement channels (quantum-to-classical maps), *all* standard divergences yield the same retrodiction: the Petz map.

In this paper, we show that the τ framework provides a precise, quantitative characterization of collapse that integrates all three results. We define *operational collapse* as $\tau \to 1$: the Petz recovery map—the unique Bayesian retrodiction functor [16]—fails completely, certifying that no physical protocol can recover the prior superposition. We derive $\tau(t)$ explicitly for the gravitational dephasing channel of Pikovski et al. [15] [Sec.], connect internal and spatial decoherence via an entanglement complementarity bridge [Sec.], establish a quantitative link between irreversibility and objectivity (the τ-R bridge) [Sec.], show that Penrose's E_G is subsumed as a special case giving only one point on the full $\tau(t)$ curve [Sec.], develop the AND→OR transition as the operational meaning of collapse [Sec.], and tabulate predictions for current and proposed experiments [Sec.].

Each result is labeled [KNOWN], [NEW SYNTHESIS], [NEW FORMULA], or [NEW BOUND]. Attribution to Liu et al., Jean et al., and Kuang et al. is marked explicitly throughout.

GRAVITATIONAL DEPHASING AS A QUANTUM CHANNEL

Pikovski Hamiltonian [KNOWN].—Consider a massive particle in a spatial superposition $|\Psi(0)\rangle = (\alpha|x_1\rangle + \beta|x_2\rangle) \otimes |\chi_{\rm int}\rangle$, where $|\chi_{\rm int}\rangle$ is the internal state. In the post-Newtonian approximation, the total Hamiltonian is [15]

$$H = \frac{p^2}{2m} + H_{\rm int}\left(1 + \frac{\Phi(\hat{x})}{c^2}\right), \tag{2}$$

where $\Phi(\hat{x})$ is the Newtonian gravitational potential evaluated at the center-of-mass position. Time evolution under Eq. (2) produces [15]

$$|\Psi(t)\rangle = \alpha|x_1\rangle \otimes U_1|\chi\rangle + \beta|x_2\rangle \otimes U_2|\chi\rangle, \tag{3}$$

where $U_j = \exp(-iH_{\rm int}(1+\Phi_j/c^2)\,t/\hbar)$ with $\Phi_j \equiv \Phi(x_j)$.

Decoherence factor [KNOWN].—Tracing over internal degrees of freedom yields the reduced spatial state

$$\rho_S(t) = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* D(t) \\ \alpha^*\beta\, D^*(t) & |\beta|^2 \end{pmatrix}, \tag{4}$$

with decoherence factor

$$D(t) = \langle\chi|\exp\left(\frac{iH_{\rm int}\Delta\Phi\, t}{\hbar c^2}\right)|\chi\rangle = \sum_n |c_n|^2 e^{-iE_n\Delta\Phi\, t/(\hbar c^2)}, \tag{5}$$

where $\Delta\Phi = \Phi(x_1) - \Phi(x_2)$, $\{E_n\}$ are internal energy eigenvalues, and $c_n = \langle E_n|\chi\rangle$.

CPTP dephasing channel [KNOWN].—The map $\mathcal{N}_{\rm grav} : \rho_S(0) \mapsto \rho_S(t)$ is a completely positive, trace-preserving (CPTP) dephasing channel with Kraus operators [8]

$$K_0 = \sqrt{\frac{1+p}{2}}\, I, \qquad K_1 = \sqrt{\frac{1-p}{2}}\, \sigma_z, \tag{6}$$

where $p(t) = |D(t)|$ and $\sigma_z = |x_1\rangle\langle x_1| - |x_2\rangle\langle x_2|$.

Many internal modes [KNOWN].—For N internal modes in a thermal state at temperature T, the decoherence factor factorizes: $D(t) = \prod_{k=1}^N d_k(t)$, where each mode contributes $d_k(t) = 1 - \varepsilon_k(t)$ with $\varepsilon_k \ll 1$. The product yields

$$|D(t)| \approx \exp\left(-\sum_k \varepsilon_k\right) = \exp(-N\langle\varepsilon\rangle), \tag{7}$$

giving Gaussian decay $p(t) = \exp(-\Gamma t^2)$ with $\Gamma = (\Delta E)^2(\Delta\Phi)^2/(2\hbar^2 c^4)$ for a thermal internal state with energy variance $(\Delta E)^2$ [15].

Temporal asymmetry parameter [NEW SYNTHESIS].—The Petz recovery map for the dephasing channel with maximally mixed reference state is self-adjoint: $\mathcal{R} = \mathcal{N}_{\rm grav}$. Hence the recovery fidelity (worst case, $|+\rangle$ input) is $F_{\rm Petz} = (1+p^2)/2$. Using the universal (rotated) Petz map [4] and defining $\tau(t) = 1 - F(t)$, we obtain

$$\boxed{\tau_{\rm grav}(t) = 1 - \exp\big(-N\Gamma t^2\big)} \tag{8}$$

for N internal modes in the large-N limit. This is a monotone from $\tau = 0$ (full coherence) to $\tau \to 1$ (operational collapse), with characteristic timescale

$$t_{\rm dec} = \frac{1}{\sqrt{N\Gamma}} = \frac{\hbar c^2\sqrt{2}}{\sqrt{N}\,\Delta E\,|\Delta\Phi|}. \tag{9}$$

Entropy production [KNOWN].—The entropy production of the gravitational dephasing channel is $\Sigma(t) = H_{\rm bin}((1+p(t))/2)$, where $H_{\rm bin}(x) = -x\log x - (1-x)\log(1-x)$ is the binary entropy. The bound $\tau \le 1 - \exp(-\Sigma/2)$ from Ref. [1] is satisfied identically. Note that Σ here is the JRSWW entropy production of the dephasing channel, $\Sigma = H_{\rm bin}((1+p)/2)$, not the linear identification $E_G t/\hbar$ used in Sec. . The two coincide at leading order for small dephasing but differ at higher orders.

INTERNAL–SPATIAL BRIDGE

[NEW SYNTHESIS] The state (3) is a pure bipartite state on $\mathcal{H}_{\rm spatial} \otimes \mathcal{H}_{\rm internal}$. Partial traces define two dephasing channels: $\mathcal{N}_S = {\rm Tr}_{\rm int}$ (yielding ρ_S) and $\mathcal{N}_I = {\rm Tr}_{\rm spatial}$ (yielding ρ_I). Define τ_S and τ_I as the temporal asymmetry parameters for each channel, evaluated for the worst-case input (equal superposition).

For the spatial channel, $\tau_S = (1 - |D(t)|)/2$. For the internal channel, $\tau_I = (1 - |D(t)|^2)/2$. Eliminating $|D|$ yields the *bridge relation*:

$$\boxed{\tau_I = 2\,\tau_S\,(1 - \tau_S).} \tag{10}$$

This is a concave parabola with maximum $\tau_I = 1/2$ at $\tau_S = 1/2$, reflecting the fact that a qubit dephasing channel saturates at $\tau = 1/2$.

Entanglement entropy [KNOWN].—Since the total state is pure, the von Neumann entropy of either subsystem equals the entanglement entropy:

$$S_{\rm ent}(t) = H_{\rm bin}\big(1 - \tau_S(t)\big). \tag{11}$$

The entropy production of both channels equals the entanglement entropy:

$$\Sigma_{\rm int} = \Sigma_{\rm spatial} = S_{\rm ent}. \tag{12}$$

This identity [NEW SYNTHESIS] follows because both channels arise from tracing out the complementary subsystem of a pure state.

Englert–Greenberger complementarity [NEW SYNTHESIS].—The wave-particle duality relation of Englert and Greenberger [20, 21] takes the form $\mathcal{V}^2 + \mathcal{D}_{\rm wp}^2 \le 1$, where $\mathcal{V}$ is visibility and $\mathcal{D}_{\rm wp}$ is which-path distinguishability. In the τ language:

$$\mathcal{V} = 1 - 2\tau_S, \qquad \mathcal{D}_{\rm wp} = \sqrt{2\tau_I}. \tag{13}$$

The duality relation becomes $\mathcal{V}^2 + \mathcal{D}^2_{\rm wp} = (1-2\tau_S)^2 + 2\tau_I = 1$, which follows identically from Eq. (10).

Pointer basis from the equivalence principle [NEW SYNTHESIS].—The equivalence principle dictates that $\Phi(\hat{x})$ in Eq. (2) couples to *position*: the interaction is diagonal in position, so by Zurek's einselection criterion [6], gravity universally selects position as the pointer basis. Unlike electromagnetic or nuclear decoherence, gravitational einselection applies to any massive system regardless of internal composition.

FROM DECOHERENCE TO OBJECTIVITY: THE τ-R BRIDGE

The previous sections established that $\tau \to 1$ certifies irreversibility. We now show that $\tau \to 1$ and objectivity (in the sense of quantum Darwinism [17]) emerge from the *same* physical parameter.

Tensor product amplification [KNOWN].—The decoherence factor (7) is a product over N independent modes. Each mode records which-path information $\varepsilon_k \ll 1$, but the product decays exponentially: $|D| \approx \exp(-N\langle\varepsilon\rangle)$. Even for $\langle\varepsilon\rangle \sim 10^{-20}$, the factor $N \sim 10^{23}$ of a macroscopic object drives $|D| \to 0$ and $\tau \to 1$.

Quantum Darwinism redundancy [KNOWN].—The redundancy R counts the number of independent environmental fragments that each contain sufficient information to determine the system's state. For the gravitational dephasing channel with N internal modes, a fragment of size $f \cdot N$ modes gives partial decoherence factor $|D_f| = \exp(-fN\Gamma_{\rm single}t^2)$, which becomes negligible when $f > f_c = 1/(\Gamma t^2)$. The redundancy is [17, 18]

$$R(t) = N\big[1 - \exp(-\Gamma_{\rm single}t^2)\big]\,. \tag{14}$$

The τ-R bridge [NEW FORMULA].—Comparing Eqs. (8) and (14), both are governed by the same dephasing parameter $\Gamma_{\rm single}$:

$$\boxed{\begin{aligned}\tau(t) &= 1 - \exp\big(-N\Gamma_{\rm single}t^2\big),\\ R(t) &= N\big[1 - \exp(-\Gamma_{\rm single}t^2)\big]\,.\end{aligned}} \tag{15}$$

At the decoherence time $t_{\rm dec} = 1/\sqrt{N\Gamma_{\rm single}}$, we have $\tau(t_{\rm dec}) = 1 - 1/e \approx 0.63$ and $R(t_{\rm dec}) \approx 1$ for $N \gg 1$. This demonstrates that *collapse ($\tau \to 1$) and objectivity ($R \to N$) occur at the same timescale.*

Spectrum broadcast structure [NEW SYNTHESIS].—Korbicz et al. [19] proved that any von Neumann-type interaction with $N \gg 1$ sub-environments generically produces spectrum broadcast structure (SBS):

$$\rho_{SE}(t) = \sum_i p_i\, |i\rangle\langle i|_S \otimes \rho_i^{(1)} \otimes \cdots \otimes \rho_i^{(N)}, \tag{16}$$

where $\{|i\rangle\}$ are the pointer states. The gravitational dephasing interaction (2) produces SBS with pointer states $\{|x_1\rangle, |x_2\rangle\}$, ensuring objective existence [19].

$\tau = 1$ and operational superselection [NEW SYNTHESIS].—When $\tau = 1$, no observable in the accessible algebra can distinguish the state from a classical mixture. This is operationally equivalent to a superselection rule on position [22].

For gravitational decoherence, two results strengthen this. (i) Danielson, Satishchandran, and Wald [23] showed that superpositions of masses radiate soft gravitons crossing the cosmological horizon—information is causally irretrievable. (ii) Kiefer and Joos [24] proved that with infinitely many graviton modes, the decoherence is permanent. Together with infrared superselection [25], $\tau = 1$ from gravitational dephasing may be exact rather than approximate.

PENROSE AS A SPECIAL CASE

The Diósi–Penrose proposal [KNOWN].—Diósi and Penrose independently proposed that the gravitational self-energy,

$$E_G = G\int d^3\mathbf{x}\, d^3\mathbf{x}'\, \frac{[\rho_1(\mathbf{x}) - \rho_2(\mathbf{x})][\rho_1(\mathbf{x}') - \rho_2(\mathbf{x}')]}{|\mathbf{x} - \mathbf{x}'|}, \tag{17}$$

sets a fundamental collapse rate $\Lambda = E_G/\hbar$ [9, 10]. Penrose's argument [11] rests on the principle that superpositions of distinct spacetime geometries are fundamentally unstable—"objective reduction" (OR).

τ bound from Penrose–Diósi entropy production [NEW BOUND].—The Diósi–Penrose model posits exponential decoherence $|D(t)| = \exp(-E_G t/\hbar)$. Artini et al. [27] computed the entropy production rate for the DP model. Following their identification $\Sigma_{\rm DP}(t) = E_G t/\hbar$, the JRSWW bound gives the *upper bound*

$$\boxed{\tau(t) \;\le\; 1 - \exp\!\left(-\frac{E_G\, t}{2\hbar}\right).} \tag{18}$$

The actual $\tau(t)$ depends on the specific CPTP map implementing the DP dynamics; the bound constrains any such implementation. At Penrose's timescale $t_P = \hbar/E_G$: $\tau(t_P) \le 1 - e^{-1/2} \approx 0.39$.

Table I summarizes the structural differences.

TABLE I. Penrose versus τ framework.

	Penrose	**τ framework**
Output	One time t_P	Full curve $\tau(t)$
Proof	Heuristic	Theorem (JRSWW [4])
Scope	Gravity only	Any CPTP map
New physics?	Yes (OR)	No (standard QM)
$\tau = 0$?	Still collapses	No collapse (theorem)

Artini et al. connection [NEW SYNTHESIS].—Artini, Lo Monaco, Donadi, and Paternostro [27] computed

the entropy production rate $\Pi(t)$ for the Diósi–Penrose model using the Wigner-function Fokker–Planck equation, finding that the frictionless DP model produces dynamics consistent with the second law only for an infinite-temperature noise field, while the dissipative extension achieves thermalization. Their entropy production Σ_{Artini} is computed from the Kullback–Leibler divergence between the Wigner function and the thermal equilibrium distribution [Eq. (2) of Ref. [27]]—the same quantity that, through the JRSWW bound, constrains our recovery fidelity via $F \geq \exp(-\Sigma/2)$. They did not use Petz recovery or compute recovery fidelity. Our contribution is to close this gap: connecting their Σ to the Petz bound gives the first thermodynamic constraint on spatial recoverability within the DP model.

Honest limitations.—Three caveats are essential.

(i) The Pikovski mechanism [15] (gravitational time dilation coupling to internal DOF) and the Diósi–Penrose mechanism [9, 10] (gravitational self-energy of the superposition) are *different physical processes.* They coincide only when specific parameter relations hold. The τ framework accommodates both as inputs but does not derive E_G from first principles.

(ii) E_G is an input parameter, not an output. The framework provides the operational definition of collapse ($\tau \to 1$) and the thermodynamic bound ($F \geq \exp(-\Sigma/2)$), but it does not predict the decoherence rate; it takes the rate as given and quantifies its consequences.

(iii) Galley, Giacomini, and Selby [26] proved that any consistent coupling between classical gravity and quantum matter is *fundamentally irreversible*—no Stinespring dilation exists. If gravity has classical aspects, then $\tau > 0$ from gravitational dephasing is genuine (not FAPP). If gravity is fully quantum, $\tau > 0$ is FAPP—but operationally indistinguishable in all current experiments.

THE AND–OR TRANSITION

[NEW SYNTHESIS, building on Liu, Bai, and Scarani [12]]

The density matrix $\rho_S = I/2$ admits two fundamentally different prior beliefs (in the retrodictive sense of Ref. [16]):

(a) *Proper mixture* (OR state): the system was in $|0\rangle$ OR $|1\rangle$ with equal probability. The prior is $\beta_1 = \frac{1}{2}(|0\rangle\langle 0|_S \otimes |\underline{0}\rangle\langle\underline{0}|_R + |1\rangle\langle 1|_S \otimes |\underline{1}\rangle\langle\underline{1}|_R)$, where R is a classical register. Liu et al. [12] showed that upon observing $|0\rangle$, the Petz retrodiction updates the belief: the system was in $|0\rangle$ with certainty. A definite past exists, and observation refines it.

(b) *Improper mixture* (AND state): the system is half of a Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. The prior is $\beta_2 = |\Phi^+\rangle\langle\Phi^+|_{SR}$. Upon observing $|0\rangle$, the Petz retrodiction returns $I/2$—no update. No definite past exists; the system was in $|0\rangle$ *and* $|1\rangle$ simultaneously. The belief is immune to evidence because it is already maximal (pure on the global system).

Decoherence as AND→OR [NEW SYNTHESIS].—The gravitational dephasing channel converts an improper mixture (AND) to a state that increasingly resembles a proper mixture (OR) as τ grows. Specifically, as $\tau \to 1$:

1. The off-diagonal elements of ρ_S vanish ($D(t) \to 0$).
2. The entanglement between spatial and internal DOF becomes maximal ($S_{\text{ent}} \to \log 2$).
3. The internal degrees of freedom carry complete which-path information (distinguishability $\mathcal{D}_{\text{wp}} \to 1$).
4. The retrodictive structure transitions from "no definite past exists" to "a definite past exists and can be inferred."

At the operational level, the AND→OR transition *is* collapse. The parameter τ quantifies its progress:

$$\begin{aligned} \tau = 0: &\quad \text{pure AND (no definite past)}, \\ 0 < \tau < 1: &\quad \text{partial transition}, \\ \tau = 1: &\quad \text{pure OR (definite past exists)}. \end{aligned} \tag{19}$$

Divergence-independence for measurements (Kuang, Torii, Buscemi [14]).—A potential objection is that τ depends on the choice of Uhlmann fidelity. Kuang et al. proved that for measurement channels—quantum-to-classical maps of the form $\mathcal{M}(\gamma_Q) = \sum_x \mathrm{Tr}\{M_x\,\gamma_Q\}\,|x\rangle\langle x|$—*all* standard f-divergences (Umegaki, Petz–Rényi, sandwiched Rényi, geometric Rényi, Belavkin–Staszewski, trace distance) yield the same retrodiction map. The unique optimizer is

$$\hat{\gamma}_{Q|x} = \frac{\sqrt{\gamma_Q}\,M_x\,\sqrt{\gamma_Q}}{\mathrm{Tr}\{M_x\gamma_Q\}}, \tag{20}$$

which is the Petz transpose output. The mathematical reason is the semiclassical (direct-sum) structure of measurement channels, which allows the divergence to decompose into independent conditional terms.

For the collapse problem, this means the AND→OR transition—diagnosed by any of these divergences—yields the same retrodiction. The measure $\tau = 1 - F_{\text{Petz}}$ is not one choice among many; it is *the* canonical irreversibility measure for measurement channels.

$\tau = 0$ AND P-CTC EQUIVALENCE

[NEW SYNTHESIS, building on Jean, Silva, and Vilasini [13]]

Jean et al. proved a bidirectional operational equivalence: every (possibly mixed) multi-time state (MTS)

has an operationally equivalent P-CTC-assisted comb, and vice versa (Theorem 1.1 of Ref. [13]). P-CTCs—post-selected closed timelike curves in the sense of Lloyd et al. [30]—allow cyclic causal influences through post-selected teleportation on maximally entangled states.

In the τ language, this yields:

$$\tau = 0 \iff \text{time-symmetric} \iff \text{P-CTC-equivalent}. \tag{21}$$

A closed system with $\tau = 0$ is not merely "reversible" in the colloquial sense—it is operationally equivalent to a process with bidirectional information flow. The "retro-causal" appearance of the quantum eraser [28, 29] is not mysterious—it is the expected behavior of any time-symmetric ($\tau = 0$) process.

Jean et al. establish a binary equivalence: a process either has P-CTC structure or it does not. What τ adds is the *continuous interpolation*: the bound $F \geq \exp(-\Sigma/2)$ quantifies how the P-CTC structure degrades as the system opens ($\tau > 0$). The transition from $\tau = 0$ to $\tau > 0$ is the loss of time-symmetric structure—the emergence of a preferred temporal direction.

For the gravitational dephasing channel specifically, the loss of P-CTC structure is controlled by the entropy production: $\tau(t) \leq 1 - \exp(-\Sigma(t)/2)$ traces the continuous degradation from $\tau = 0$ (P-CTC-equivalent, no collapse) to $\tau \to 1$ (no time-symmetry, full collapse).

WHY GRAVITY CAUSES COLLAPSE: THE CAUSAL CHAIN

[NEW SYNTHESIS] Each step in the following chain has explicit mathematical justification:

$$\begin{aligned}
&\Phi(x_1) \neq \Phi(x_2) \\
&\quad \xrightarrow{\text{Eq. (2)}} U_1 \neq U_2 \text{ (different internal evolution)} \\
&\quad \xrightarrow{\text{Eq. (5)}} D(t) \to 0 \text{ (internal DOF distinguish paths)} \\
&\quad \xrightarrow{\text{Eq. (12)}} \Sigma > 0 \text{ (entropy production)} \\
&\quad \xrightarrow{\text{Eq. (8)}} \tau > 0 \text{ (retrodiction fails)} \\
&\quad \xrightarrow{\text{Liu et al. [12]}} \text{AND} \to \text{OR (definite past emerges)}.
\end{aligned} \tag{22}$$

The root cause is the equivalence principle: $\Phi(\hat{x})$ couples to position, creating position-dependent phase accumulation in the internal degrees of freedom. The non-uniformity of the gravitational redshift—the fact that clocks at different heights tick at different rates—is what drives the entire chain.

Gravity occupies a unique position among decoherence mechanisms: (i) it couples universally to all forms of energy-momentum; (ii) it cannot be shielded [9, 15]; (iii) the equivalence principle guarantees position as the pointer basis. Consequently, τ_{grav} constitutes the *irreducible floor* of temporal asymmetry: for any massive system in a spatial superposition, $\tau_{\text{total}} \geq \tau_{\text{grav}} > 0$.

EXPERIMENTAL PREDICTIONS

[NEW SYNTHESIS] Table II collects the gravitational temporal asymmetry τ_{grav} for major experiments.

TABLE II. Gravitational temporal asymmetry τ_{grav} for current and proposed experiments. $t_{\text{DP}} = \hbar/E_G$ is the Diósi–Penrose timescale. All current experiments have $\tau_{\text{grav}} \ll 1$.

Experiment	m (kg)	t_{DP} (s)	τ_{grav}
Fein 2019 [32]	4×10^{-23}	$\sim 10^{14}$	$\ll 10^{-14}$
Arndt 2025 [33]	3×10^{-22}	$\sim 10^{12}$	$\ll 10^{-12}$
Delić 2020 [34]	3×10^{-18}	$\sim 10^{5}$	$\ll 10^{-5}$
Panda 2024 [35]	2×10^{-25}	$\sim 10^{21}$	$\ll 10^{-19}$
BMV (proposed) [36]	10^{-14}	~ 15	~ 0.004
Tagg mirror [9]	2×10^{-4}	$\sim 1.5\ \mu$s	~ 1

Universal bound on spatial recoverability [NEW SYNTHESIS].—From the master inequality chain of Ref. [1], any gravitational dephasing channel satisfies

$$F_{\text{spatial}}(t) \geq \exp\big(-\Sigma_{\text{grav}}(t)/2\big). \tag{23}$$

This constrains the performance of *any* recovery protocol: spatial coherence cannot be restored beyond the limit set by gravitational entropy production.

$\tau(t)$ as baseline; deviations signal new physics [NEW FORMULA].—The standard-QM prediction is $\tau_{\text{baseline}}(t) = 1 - \exp(-\Sigma(t)/2)$. If the DP objective reduction mechanism is correct, the observed $\tau(t)$ would match $1 - \exp(-E_G t/2\hbar)$. Any deviation—$\tau_{\text{obs}}(t) > \tau_{\text{baseline}}(t)$—would indicate collapse faster than standard QM predicts, pointing to new physics beyond unitary quantum mechanics. Conversely, exact agreement with the baseline at all t would disfavor objective collapse models.

Gran Sasso constraints [KNOWN].—Donadi et al. [38] tested the DP model underground using spontaneous radiation from germanium. The parameter-free Diósi model is ruled out ($R_0 > 0.54$ Å), but Penrose's version (with nuclear-scale smearing) remains viable.

BMV experiment [KNOWN].—The Bose–Marletto–Vedral experiment [36, 37] proposes to detect gravitationally-induced entanglement. In our framework, this corresponds to $\tau_{\text{grav}} \sim 0.004$ (for $t = 1$ s, $m = 10^{-14}$ kg), where partial decoherence is measurable but collapse is far from complete.

Existing Petz recovery experiments.—Png and Scarani [39] implemented Petz recovery on an ion trap processor; Singh et al. [40] on NMR. Re-analysis of their recovery fidelities against the channel entropy production can directly test the bound (23).

DISCUSSION

We have shown that wavefunction collapse—the transition from quantum superposition to classical definiteness—admits a precise information-theoretic characterization: *operational collapse is Petz recovery failure* ($\tau \to 1$). Three recent results give this characterization concrete physical content:

(i) *Liu et al. [12]: Collapse is the AND→OR transition.* The same density matrix $\rho = I/2$ gives entirely different retrodictions depending on whether the mixture is proper (OR) or improper (AND). Decoherence converts improper to proper, changing what retrodiction means—not just how well it works. The τ parameter tracks this transition continuously from $\tau = 0$ (pure AND, no definite past) to $\tau = 1$ (pure OR, definite past exists).

(ii) *Jean et al. [13]: $\tau = 0 \Leftrightarrow$ P-CTC-equivalent.* A closed system has the operational structure of a time machine. Collapse ($\tau \to 1$) is the destruction of this structure. The τ framework provides the continuous interpolation that Jean et al.'s binary theorem lacks.

(iii) *Kuang et al. [14]: τ is unique for measurements.* All standard divergences yield the same retrodiction for measurement channels. The AND→OR transition is diagnosed identically regardless of which divergence is used.

The causal chain (22) traces collapse from its physical origin (non-uniform gravitational redshift) through the mathematical machinery (entropy production, Petz recovery bound) to its operational consequence (the emergence of a definite past). The Penrose timescale $t_P = \hbar/E_G$ is recovered as a special case [Eq. (18)], but the τ framework provides the full dynamics, the thermodynamic bound, and the connection to objectivity—all within standard quantum mechanics.

What we answer.—(i) *When does collapse happen?* When $\tau \to 1$, quantified by Eq. (8). (ii) *Why operationally only one outcome?* SBS ensures all observers agree [19]; $\tau = 1$ certifies no recovery protocol exists. (iii) *What is the timescale?* $t_{\rm dec} = 1/\sqrt{N\Gamma_{\rm single}}$; under Diósi–Penrose, $t_c = \hbar/E_G$.

What we do not answer.—The *ontological* question (whether other branches "cease to exist") lies outside the framework. Our definition of collapse is operational. The distinction between operational and ontological collapse is a choice of interpretation. Nor do we explain *why* only one outcome occurs—the Born rule is assumed, not derived.

Alternative mathematical route.—The recovery bound $F \geq \exp(-\Sigma/2)$ requires identifying a CPTP map whose entropy production equals Σ. An alternative approach via quantum hypothesis testing [46] defines $\Sigma = D(\rho(z_1)\|\rho(z_2))$ directly from quantum states at different spacetime points, bypassing the need for an explicit channel. This may strengthen the gravitational applications in Paper II [43].

Open questions.—(i) The intermediate regime $0 < \tau < 1$ deserves richer operational characterization. (ii) Observer dependence: different system-environment partitions assign different τ. (iii) Isolating $\tau_{\rm grav}$ experimentally requires extraordinary mass scales or isolation. (iv) Non-Markovian extensions via the process tensor framework [31] may extend the equivalence chain to settings where Σ is transiently negative. (v) The scope of claims: this framework provides a quantitative language for collapse and a universal bound, not a resolution of the measurement problem.

Outlook.—The τ-R bridge suggests a broader program. In a companion paper [43], we apply the same framework to strong gravitational fields, where $\Sigma = r_s/r$ and the Petz recovery bound constrains spacetime structure near horizons. The AND→OR transition developed here may inform quantum error correction strategies: QEC can be understood as the deliberate construction of a local environment where $\tau \approx 0$—an artificial suppression of the arrow of time, preventing the AND→OR transition that would destroy encoded quantum information.

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